

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards. (BL2, BL5)

Assessment Tool Weightage: 40%

Questions: 3 Total Marks: 30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CSA07 — COMPUTER NETWORKS

CO5 Assessment Tool 1 — Constraint-Based Problem Solving: Answers

CO5: Design network topologies using networking devices, transmission media, and cabling standards

Q1. Rural Branch Office — Connectivity, Devices, and Security on a Limited Budget

Constraints to satisfy simultaneously: no wired high-speed internet available, must support cloud apps + video conferencing + secure HQ communication, limited budget, and uninterrupted operations required.

Problem Decomposition

- Sub-problem 1 — how to get internet access at all without wired broadband
- Sub-problem 2 — how to keep that connection reliable enough for "uninterrupted business operations"
- Sub-problem 3 — how to secure the link back to headquarters and the local office network
- Sub-problem 4 — how to keep all of this affordable now while leaving room to scale later

Recommended Solution

Requirement	Recommendation
Primary internet	Satellite broadband (e.g. VSAT/LEO satellite such as Starlink Business) as the primary WAN link — the only realistic option with no wired infrastructure nearby
Backup/failover link	4G/5G cellular router as a secondary WAN, in an SD-WAN dual-WAN failover setup
Core device	SD-WAN capable router/firewall with dual-WAN failover and QoS support
Security gateway	Unified Threat Management (UTM) firewall combining firewall + IPS + antivirus at the office edge
HQ connectivity	Site-to-site IPsec VPN tunnel from the branch router to headquarters
Local network	A simple star-topology LAN (switch + Wi-Fi AP) for office devices

Justification Against Each Constraint

- Bandwidth limitations: QoS rules on the router prioritize video-conferencing and VPN traffic over lower-priority traffic (e.g. bulk downloads), and cloud apps are latency-tolerant enough to work over satellite/LTE if prioritized correctly.
- Reliability: the dual-WAN (satellite + cellular) failover setup means if one link degrades or drops, the SD-WAN router automatically shifts traffic to the other — directly satisfying the "uninterrupted operations" requirement without needing an expensive dedicated leased line.
- Cost: satellite/cellular links and an SD-WAN router are far cheaper than trenching fiber or leasing a dedicated circuit to a remote rural site, which may not even be physically available at any price.
- Future scalability: SD-WAN policies and bandwidth can be increased purely through plan/subscription upgrades or adding a third WAN link, with no need to re-architect the branch network as usage grows.

Security Mechanisms

- Site-to-site IPsec VPN for all HQ-bound traffic, so data never traverses the satellite/cellular links unencrypted.
- UTM firewall with intrusion prevention at the branch edge, and endpoint protection plus multi-factor authentication on employee devices/cloud logins.

Q2. Campus-Wide Wireless Network for 3,000+ Students on a Limited AP Budget

Constraints: very large simultaneous user count across varied spaces (classrooms, labs, libraries, hostels), a limited number of access points affordable, and interference/security must still be handled properly.

Problem Decomposition

- Sub-problem 1 — how to serve many simultaneous users per AP, since AP count is capped by budget
- Sub-problem 2 — where to physically place the limited APs across very different space types
- Sub-problem 3 — how to choose frequency/channels to minimize interference
- Sub-problem 4 — how to authenticate thousands of users securely

Recommended Design

- Access point standard: Wi-Fi 6 (802.11ax) APs — OFDMA and MU-MIMO let each AP serve many simultaneous clients efficiently, meaning fewer APs are needed to cover the same user density compared to older 802.11n/ac hardware.
- Placement: use a proper site survey to place APs centrally (ceiling-mounted) in each space; deploy higher AP density in high-traffic zones (library, hostel common areas, large lecture halls) and lighter coverage in smaller classrooms, rather than spreading the limited budget evenly and under-serving busy areas.
- Frequency band: dual-band operation — 5 GHz prioritized for high-density areas (more non-overlapping channels, less interference, supports more concurrent high-throughput users) with 2.4 GHz kept available for wider coverage and older/legacy student devices.
- Architecture: centralized Wireless LAN Controller (WLC) managing all APs, so channel assignment and transmit power are coordinated network-wide to minimize co-channel interference between adjacent APs — something unmanaged, independently-configured APs cannot do well at this scale.
- Authentication: WPA3-Enterprise with 802.1X, tied to the university's existing student directory (RADIUS server) so each student authenticates with their own credentials; a separate captive-portal guest VLAN keeps visitor traffic isolated from the academic network.

Justification — Balancing Coverage, Performance, Security, and Cost

- Coverage vs. cost: concentrating the limited AP budget where user density is actually highest (rather than spreading evenly) gets more effective coverage per rupee spent.
- Performance: Wi-Fi 6's higher per-AP client capacity directly compensates for having fewer total APs than an ideal unconstrained deployment would use.
- Security: WPA3-Enterprise plus per-user RADIUS authentication (rather than one shared password) gives individual accountability and lets compromised credentials be revoked without affecting every other student.
- Interference: centralized WLC channel planning plus deliberate 5 GHz/2.4 GHz split keeps the dense AP layout from degrading its own performance through self-interference.

Q3. Financial Institution — Secure Network Architecture Upgrade

Constraints: must strengthen cybersecurity for sensitive customer data and online banking access, comply with strict regulatory security policy, and do so without materially hurting performance or blowing the budget.

Problem Decomposition

- Sub-problem 1 — how to segment the network so a breach in one area cannot reach sensitive systems
- Sub-problem 2 — where to place firewalls and intrusion detection for maximum effect without adding unnecessary cost/latency everywhere

- Sub-problem 3 — how to authenticate employees strongly without slowing down daily operations
- Sub-problem 4 — how to satisfy compliance requirements (audit trails, encryption) within budget

Recommended Architecture

- Network segmentation: separate VLANs for customer-data/core banking servers, employee workstations, the online-banking-facing DMZ, and guest/visitor access — so compromise of one segment does not automatically expose the others.
- Micro-segmentation for the most sensitive assets: the customer-database servers sit behind their own internal firewall rules (a zero-trust-style policy) even from other internal VLANs, not just from the outside.
- Firewall deployment: a next-generation firewall (NGFW) with integrated IPS at the internet/DMZ boundary, plus internal firewalling between VLANs — tiered so the strongest, most expensive controls protect only the most sensitive segments rather than being applied uniformly everywhere.
- Authentication: multi-factor authentication (MFA) required for any employee access to customer data or online-banking systems, role-based access control (RBAC) so staff only reach what their role needs, and 802.1X port-based authentication on network switches to stop unauthorized devices from joining the internal LAN at all.
- Intrusion detection: network-based IDS/IPS monitoring at the DMZ and core boundaries, host-based IDS on the critical database/application servers, feeding a SIEM system for centralized log correlation and the audit trail regulators require.

Justification Against Each Constraint

- Security: segmentation + micro-segmentation + MFA + IDS/IPS together give defense-in-depth, so no single control failure exposes customer data directly.
- Performance: tiering controls (heaviest inspection only at the DMZ boundary and around the most sensitive VLAN, lighter internal firewalling elsewhere) avoids forcing every internal packet through the most expensive inspection path, keeping day-to-day performance acceptable.
- Compliance: SIEM-based logging, encryption in transit (TLS) and at rest, and documented RBAC/MFA policies map directly to the audit-trail and access-control evidence financial regulators require.
- Budget: reusing existing switch hardware that already supports VLANs and 802.1X (rather than a full infrastructure replacement), and concentrating the most expensive controls (NGFW, micro-segmentation) only around the highest-value segments, keeps the upgrade cost proportionate rather than maximizing spend everywhere uniformly.